A Norming Family of Uniform Numerical Bollobas Operators

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Source and Target

The source paper is S. Dantas, M. Jung, and O. Roldan, *Norm attaining operators which satisfy a Bollobas type theorem*, arXiv:1910.05726. After Proposition 2.8, the authors construct compact operators $T \in \mathcal{A}_\nu(H)$, where H is a separable infinite-dimensional real Hilbert space, such that

$$1 = \nu(T) < \|T\|,$$

and such that the Bollobas modulus in the definition of $\mathcal{A}_\nu(H)$ can be chosen independently of the particular operator T. They write:

> Let us recall that in Corollary 2.6, we proved that if a compact operator T defined on a Hilbert space is such that $\nu(T) = \|T\| = 1$, then T must belong to the set $\mathcal{A}_{\text{nu}}(H)$. However, the following result (inspired by [2, Example 1.9]) provides us a wide class of operators $T \in \mathcal{A}_{\text{nu}}(H)$ such that $1 = \nu(T) < \|T\|$ and , in particular, examples of operators which belong to $\mathcal{A}_{\text{nu}}$ but not to $\mathcal{A}_{\|\cdot\|}$. Notice, by item (iii) below, that T belongs to the set $\mathcal{A}_{\text{nu}}$ in a uniform sense, that is, the η does not depend on the operator T defined there. We do not know how often this happens, that is, we do not know, for instance, whether the set of such an operators could be norming for the whole space.
>
> **Proposition 2.8.** *Let H be a separable infinite dimensional real Hilbert space. Then, there is $T \in \mathcal{L}(H)$ such that*
>
> *(i) T is a compact operator.*
> *(ii) $1 = \nu(T) < \|T\|$ and T attains its numerical radius.*
> *(iii) given $\varepsilon > 0$, there is $\eta(\varepsilon) > 0$ such that whenever $x_0 \in S_H$ satisfies*
>
> $$|\langle Tx_0, x_0\rangle| > 1 - \eta(\varepsilon),$$
>
> *there is $x_1 \in S_H$ such that $\nu(T) = \langle Tx_1, x_1\rangle = 1$ and $\|x_1 - x_0\| < \varepsilon$.*
>
> *In particular, $T \in \mathcal{A}_{\text{nu}}(H)$ and $T \notin \mathcal{A}_{\|\cdot\|}(H, H)$.*
>
> *Proof.* Let $0 < \alpha \leqslant 1$ and $\{\alpha_n\}$ be a sequence such that $|\alpha_1| > 1$, $-1 < \alpha_n < 1$ for $n \geqslant 2$, and $\alpha_n \to 0$ as $n \to \infty$. Let $\{J_1, J_2, J_3\}$ be a partition of $\mathbb{N}$ such that $|J_1| = |J_2| = \aleph_0$,

The source does not introduce a separate dual-pairing interpretation for this sentence. In the numerical-radius part of the paper, the scalar quantity attached to an operator is the quadratic form $x \mapsto \langle Tx, x\rangle$. We therefore formalize "norming for the whole space" in this source-local sense and prove that the answer is yes.

Definitions

For an operator $T \in \mathcal{L}(H)$, its numerical radius is

$$\nu(T) = \sup_{\|x\|=1} |\langle Tx, x\rangle|.$$

In a real Hilbert space, the states in the definition of $\mathcal{A}_\nu(H)$ reduce to pairs (x, x), so $T \in \mathcal{A}_\nu(H)$ with $\nu(T) = 1$ means: for every $\varepsilon > 0$ there is $\eta(\varepsilon, T) > 0$ such that whenever $x \in S_H$ and

$$|\langle Tx, x\rangle| > 1 - \eta(\varepsilon, T),$$

there is $x_1 \in S_H$ with $\langle Tx_1, x_1\rangle = 1$ and $\|x_1 - x\| < \varepsilon$.

Call a family $\mathscr{G} \subset \mathcal{L}(H)$ *numerically norming for H* if

$$\|x\|^2 = \sup_{T\in\mathscr{G}} |\langle Tx, x\rangle| \qquad (x \in H).$$

Set

$$\eta_0(\varepsilon) = \min\{\varepsilon^2/4, 1/4\} \qquad (\varepsilon > 0).$$

Let $\mathscr{U}_{\eta_0}(H)$ be the set of all compact operators $T \in \mathcal{L}(H)$ such that $1 = \nu(T) < \|T\|$, T attains its numerical radius, and the $\mathcal{A}_\nu(H)$ condition holds with the same modulus η_0.

Idea of the Proof

Proposition 2.8 uses a finite-dimensional identity block and an infinite compact skew-symmetric tail. The skew-symmetric part does not contribute to $\langle Tx, x\rangle$ over a real Hilbert space; only the identity block contributes. Thus if the identity block is F, then

$$\langle Tx, x\rangle = \|P_F x\|^2.$$

Near numerical-radius attainment forces x close to F, and normalizing $P_F x$ gives an exact numerical-radius attaining point. Since we are free to choose F, taking $F = \operatorname{span}\{x\}$ gives an operator in the source's class that norms any prescribed vector x. Since every operator in that class has numerical radius 1, no operator can exceed $\|x\|^2$ on a vector x.

Theorem

Theorem 1. *Let H be a separable infinite-dimensional real Hilbert space. Then $\mathscr{U}_{\eta_0}(H)$ is numerically norming for H:*

$$\|x\|^2 = \sup_{T\in\mathscr{U}_{\eta_0}(H)} |\langle Tx, x\rangle| \qquad (x \in H).$$

More concretely, for every nonzero finite-dimensional subspace $F \subset H$ there is $T_F \in \mathscr{U}_{\eta_0}(H)$ such that $T_F|_F = I_F$. Hence the same class even norms every finite set of vectors simultaneously after taking F to contain that set.

Proof. Fix a nonzero finite-dimensional subspace $F \subset H$. Since H is separable and infinite-dimensional, choose an orthonormal decomposition

$$H = F \oplus \bigoplus_{k=1}^{\infty} \operatorname{span}\{u_k, v_k\}.$$

Choose real scalars $\beta_1 > 1$, $0 < \beta_k < 1$ for $k \geq 2$, and $\beta_k \to 0$. Define $T_F \in \mathcal{L}(H)$ by

$$T_F f = f \quad (f \in F), \qquad T_F u_k = -\beta_k v_k, \qquad T_F v_k = \beta_k u_k.$$

The operator is compact: it is the direct sum of the identity on finite-dimensional F and 2×2 blocks whose norms β_k tend to zero. Also

$$\|T_F\| = \max\{1, \sup_k \beta_k\} = \beta_1 > 1.$$

Let $x = f + w$, where $f \in F$ and $w \in F^\perp$. The block operator on $F^\perp$ is skew-symmetric, hence

$$\langle T_F x, x\rangle = \|f\|^2 + \langle T_F w, w\rangle = \|f\|^2.$$

Therefore $\nu(T_F) \leq 1$, and equality holds on every unit vector of F. Thus $\nu(T_F) = 1$, and T_F attains its numerical radius.

We next verify the uniform $\mathcal{A}_\nu(H)$ estimate. Let $0 < \varepsilon \leq 1$, and suppose $x \in S_H$ satisfies

$$|\langle T_F x, x\rangle| > 1 - \varepsilon^2/4.$$

Writing $x = f + w$ as above, we get

$$\|f\|^2 > 1 - \varepsilon^2/4, \qquad \|w\| < \varepsilon/2.$$

Set $x_1 = f/\|f\| \in S_F$. Then $\langle T_F x_1, x_1\rangle = 1$, and

$$\|x - x_1\| \leq \|w\| + |1 - \|f\|| < \varepsilon/2 + \varepsilon^2/4 < \varepsilon.$$

Thus the same modulus $\eta(\varepsilon) = \varepsilon^2/4$ works for $0 < \varepsilon \leq 1$. For $\varepsilon > 1$, the estimate for $\varepsilon = 1$ is stronger than required. Hence η_0 is an admissible common modulus for every T_F.

Since each T_F is compact, satisfies $1 = \nu(T_F) < \|T_F\|$, attains its numerical radius, and has common modulus η_0, we have $T_F \in \mathscr{U}_{\eta_0}(H)$. For any $T \in \mathscr{U}_{\eta_0}(H)$ and any $x \in H$,

$$|\langle Tx, x\rangle| \leq \nu(T)\|x\|^2 = \|x\|^2.$$

Conversely, if $x \neq 0$, take $F = \operatorname{span}\{x\}$. Then $T_F x = x$, so

$$|\langle T_F x, x\rangle| = \|x\|^2.$$

The equality is trivial for $x = 0$. Hence $\mathscr{U}_{\eta_0}(H)$ is numerically norming for H. □

Verification and Limitations

The proof is a direct parameterization of the source construction, with the finite identity block allowed to vary. No computational verification is relevant.

This is recorded as a full solution to the extracted source question under the source-local numerical-radius meaning of "norming":

$$\|x\|^2 = \sup_{T \in \mathscr{U}_{\eta_0}(H)} |\langle Tx, x\rangle|.$$

This is the interpretation supported by Proposition 2.8, where the relevant scalar is $\langle Tx, x\rangle$. The paper does not define an additional dual-pairing meaning in which operators would be norming for the Banach space $\mathcal{L}(H)$ itself; no claim is made about such a different, external formulation.

Bounded novelty check: the run indexes had no full packet for arXiv:1910.05726 before this upgrade. Web searches on June 23, 2026 for the source title, the phrase "We do not know how often this happens", the phrase "norming for the whole space", and related phrases involving $\mathcal{A}_\nu$, numerical radius, and norming did not reveal a later answer. The arXiv abstract page for 1910.05726 was used to confirm source metadata.

Human Review Recommendation

Check only the interpretive scope: if "norming for the whole space" is read in the numerical-radius sense used in the source paragraph, the theorem answers the question. If a reviewer wants a different operator-space dual-pairing problem, that should be split off as a separate, newly formalized question.

References

[1] S. Dantas, M. Jung, and O. Roldan, *Norm attaining operators which satisfy a Bollobas type theorem*, arXiv:1910.05726.

[2] F. F. Bonsall and J. Duncan, *Numerical Ranges of Operators on Normed Spaces and of Elements of Normed Algebras*, London Mathematical Society Lecture Note Series 2, Cambridge University Press, 1971.